# ENUMERATION OF EXTENDABLY 312-AVOIDING ALTERNATING SIGN RECTANGLES

KNOWLEDGE SENSE

Abstract. Höngesberg, Konvalinka and Linusson introduced extendable pattern avoidance for alternating sign rectangles and obtained recurrence relations for the number $S_{r,k,d}$ of $r \times k$ alternating sign rectangles with $d$ nonempty rows that are the upper left corner of some 312-avoiding alternating sign matrix. They conjectured an explicit formula for $S_{r,k,d}$. We prove this formula for all integers $r, k \geq d \geq 0$. We solve the recurrences along the diagonals $(r, k, d) = (d + a, d + b, d)$: the generating function of each diagonal is $P\big((1-t)^{-a} + R^b(1-tR)^{-a}\big)$, where $R$ is the generating function of the large Schröder numbers and $P = R/(1+R)$. Using a diagonal form of Lagrange inversion, we show that the conjectured formula has the same diagonal generating functions.

## 1. Introduction

An alternating sign matrix (ASM) is a square matrix with entries in $\{-1, 0, 1\}$ in which the nonzero entries of every row and every column alternate in sign and sum to 1. An ASM $M$ of order $N$ *contains* a permutation $\pi$ of length $m$ if there are rows $i_1 < \cdots < i_m$ and columns $j_1 < \cdots < j_m$ with $M_{i_\ell, j_{\pi(\ell)}} = 1$ for $\ell = 1, \ldots, m$, and *avoids* $\pi$ otherwise. Johansson and Linusson [4] showed that the number of 312-avoiding ASMs of order $n \geq 1$ is the large Schröder number $\mathrm{Sch}_{n-1}$; see [6].

Höngesberg, Konvalinka and Linusson [3] recently introduced alternating sign rectangles (ASRs), a rectangular generalization of ASMs, together with *extendable* pattern avoidance, in the spirit of the extended pattern avoidance of [2, 5]: an ASR extendably avoids $\pi$ if it is the upper left corner of some ASM avoiding $\pi$ (see Section 2). Let $S_{r,k,d}$ denote the number of $r \times k$ ASRs with exactly $d$ nonempty rows that extendably avoid 312. This number vanishes unless $r, k \geq d \geq 0$. For each pattern in $\{312, 132, 213, 231\}$ they established a system of recurrence relations which, together with the enumeration of [4], determines the corresponding numbers, and for $\pi = 312$ they conjectured a closed formula [3, Conjecture 4.5]. With $C_j = \frac{1}{j+1}\binom{2j}{j}$ denoting the Catalan numbers, the conjectured formula reads

$$\begin{aligned} F_{r,k,d} = {} & \sum_{i=1}^{d} 2^{i-1} \binom{k}{i} \left( \binom{r-2}{d-i} - 2\binom{r-2}{d-i-2} \right) + \binom{r}{d} \\ & + \sum_{i \geq 0} C_{2i} \binom{r-1+2i}{d-1-2i} + 2\sum_{i=1}^{d} (-1)^i \binom{r}{d-i}. \end{aligned} \tag{1.1}$$

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Binomial coefficients with a negative upper index occur here when $r \leq 1$; the convention is fixed in Section 2.2. The sum over $i \geq 0$ is finite, since its terms vanish for $2i > d-1$. Our main result confirms the conjecture.

**Theorem 1.1.** *For all integers $r, k, d$ with $r, k \geq d \geq 0$ we have $S_{r,k,d} = F_{r,k,d}$.*

In [3, p. 9] it is checked that $F$ takes the boundary values $F_{0,k,0} = F_{r,0,0} = 1$, $F_{1,k,1} = k$ and $F_{r,1,1} = r$, and, using a hypergeometric identity and a result of Barry [1], that $F_{n,n,n} = \mathrm{Sch}_{n-1}$ for $n \geq 1$. Closed formulas for $S_{r,k,d}$ on some further slices are obtained in [3, Section 5]: for $d = r$, by induction from the recurrence in terms of refined Schröder numbers and via Schröder ballot numbers [3, Theorems 5.1 and 5.2, Corollary 5.3]; for $d = k < r$ [3, Corollary 5.4]; and for $S_{r,r+2,r}$, through a bijection with Delannoy paths that do not cross the diagonal vertically [3, Theorem 5.5 and Corollary 5.6]. Theorem 1.1 covers the whole range of parameters.

Our approach is to organize the numbers $S_{r,k,d}$ by the excesses $a = r - d$ and $b = k - d$. For integers $a, b \geq 0$ put

$$T_{a,b}(t) = \sum_{d \geq 0} S_{d+a,d+b,d}\, t^d,$$

let $R(t) = \sum_{n \geq 0} \mathrm{Sch}_n t^n$, and let $P = R/(1+R)$.

**Theorem 1.2.** *For all integers $a, b \geq 0$,*

$$T_{a,b}(t) = P\left(\frac{1}{(1-t)^a} + \frac{R^b}{(1-tR)^a}\right).$$

Summing over $a$ and $b$ gives the trivariate generating function.

**Corollary 1.3.** *In $\mathbb{Q}[[x, y, t]]$,*

$$\sum_{a,b,d \geq 0} S_{d+a,d+b,d}\, x^a y^b t^d = \frac{P}{(1-y)\big(1 - x/(1-t)\big)} + \frac{P}{(1-yR)\big(1 - x/(1-tR)\big)}.$$

*In particular, this series is algebraic of degree at most two over $\mathbb{Q}(x, y, t)$.*

*Organization.* Section 2 recalls the definitions, fixes conventions and states the recurrences. In Section 3 we derive Theorem 1.2 from these recurrences: for $d < \min(r, k)$ and for $d = r < k$ they become a first-order relation between $T_{a,b}$ and $T_{a,b-1}$, which is solved in closed form, while the recurrence for $d = k < r$ expresses $T_{a,0}$ in terms of the series $T_{0,h}$, $h \geq 0$. In Section 4 we compute the generating function of $F_{r,k,d}$ in $d$ for fixed $r$ and $k$, and extract the diagonal $(d+a, d+b, d)$ by a standard form of Lagrange inversion. Comparing coefficients proves Theorem 1.1 in Section 5, where we also record some special cases; for $a = b = 0$ this gives a second proof of $F_{n,n,n} = \mathrm{Sch}_{n-1}$.

*Dependencies.* Besides elementary facts on formal power series, the proof uses the recurrences of [3, Propositions 4.2 and 4.3, Theorem 4.4] and the enumeration of 312-avoiding ASMs in [4], in the form recalled in [3, Sections 3 and 4]. We do not reprove these results. The formula (1.1) is the conjecture of [3]; what is new here is its proof.

## 2. Preliminaries

2.1. **Alternating sign rectangles.** We follow [3, Section 2]. Let $r, k \geq 0$. An *alternating sign rectangle* of size $r \times k$ is an $r \times k$ matrix $A$ with entries in $\{-1, 0, 1\}$ such that

(1) the nonzero entries alternate in sign in each row and in each column;
(2) in each row containing a nonzero entry, the leftmost and the rightmost nonzero entries are 1;
(3) in each column containing a nonzero entry, the topmost nonzero entry is 1.

A row or column without nonzero entries is *empty*. Rows and columns play different roles: a nonempty row sums to 1, while a nonempty column sums to 0 or 1. For $r = 0$ or $k = 0$ the only ASR is the empty matrix, which has no nonempty rows. An ASM is a square ASR whose rows and columns all sum to 1.

An ASR $A$ of size $r \times k$ *extendably avoids* 312 if there are an integer $N \geq \max(r, k)$ and an ASM $M$ of order $N$ that avoids 312 and whose upper left $r \times k$ submatrix is $A$. Thus $M$ is obtained from $A$ by adding rows below and columns to the right. Here $M$ contains 312 if there are rows $i_1 < i_2 < i_3$ and columns $j_1 < j_2 < j_3$ with

$$M_{i_1,j_3} = M_{i_2,j_1} = M_{i_3,j_2} = 1,$$

the remaining entries of $M$ being arbitrary. Each ASR is counted once, however many extensions it has. We write $S_{r,k,d}$ for the number of extendably 312-avoiding ASRs of size $r \times k$ with exactly $d$ nonempty rows; this is the number $S^{312}_{r,k,d}$ of [3].

The only ASR with no nonempty row is the zero matrix, and it is the upper left corner of the permutation matrix of order $r + k$ with ones in positions $(i, r + k + 1 - i)$, which avoids 312. Hence $S_{r,k,0} = 1$ for all $r, k \geq 0$, in accordance with the convention of [3, Section 4]. Extendable avoidance is more restrictive than avoidance of 312 inside the rectangle.

**Example 2.1.** We have $S_{2,2,1} = 3$. An ASR of size $2 \times 2$ with one nonempty row has exactly one nonzero entry, which equals 1; there are four such matrices, and all of them avoid 312. The matrix

$$A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

is the upper left corner of the permutation matrix of 312, but it is not the upper left corner of any 312-avoiding ASM. Indeed, in every ASM $M$ with upper left corner $A$ the first row contains a 1 in some column $j_3 \geq 3$, and the second column contains a 1 in some row $i_3 \geq 3$; together with $M_{2,1} = 1$ this is an occurrence of 312. The other three matrices are the upper left corners of the permutation matrices of 132, 231 and 321, which avoid 312; this agrees with $F_{2,2,1} = 3$.

2.2. **Formal power series and binomial coefficients.** We work with formal power series over $\mathbb{Q}$ and write $[t^n]\, A$ for the coefficient of $t^n$ in $A$. A series with constant term 1 has a unique square root with constant term 1, and $\sqrt{\cdot}$ always denotes this square root. A series $B$ is substituted into another series only when $B(0) = 0$.

For integers $u$ and $v$ we put

$$\binom{u}{v} = [z^v]\,(1+z)^u \quad (v \geq 0), \qquad \binom{u}{v} = 0 \quad (v < 0),$$

the expansion being taken at $z = 0$. For $u \geq 0$ these are the usual binomial coefficients; for instance $\binom{-1}{0} = 1$. Thus $\sum_{v\geq 0} \binom{u}{v} z^v = (1+z)^u$ for every integer $u$, and for every integer $a \geq 0$,

$$\sum_{h\geq 0} \binom{a+h-1}{h} x^h = (1-x)^{-a}. \tag{2.1}$$

Empty sums are zero. With these conventions (1.1) defines $F_{r,k,d}$ for all integers $r, k, d \geq 0$. In the range $r, k \geq d \geq 0$ of Theorem 1.1, a negative upper index with a nonnegative lower index occurs only in the term $\binom{-1}{0}$, for $r = d = 1$; the evaluation $F_{1,k,1} = k$ recorded in [3, p. 9] uses $\binom{-1}{0} = 1$.

2.3. **Schröder numbers.** The large Schröder numbers $\mathrm{Sch}_n$ begin with 1, 2, 6, 22, 90, 394, and their generating function is

$$R(t) = \sum_{n\geq 0} \mathrm{Sch}_n t^n = \frac{1 - t - \sqrt{1 - 6t + t^2}}{2t},$$

see [3, Section 3]. Equivalently, $R$ is the unique series with $R(0) = 1$ and

$$R = 1 + tR + tR^2. \tag{2.2}$$

Indeed, (2.2) determines the coefficients of $R$ recursively, and the series $1 - t - 2tR$ has constant term 1 and square $(1-t)^2 - 4t\big((1-t)R - tR^2\big) = 1 - 6t + t^2$ by (2.2), so that

$$\sqrt{1 - 6t + t^2} = 1 - t - 2tR. \tag{2.3}$$

Since $1 + R$ has constant term 2, the series $P = R/(1+R)$ is defined.

**Lemma 2.2.** *The following identities hold in $\mathbb{Q}[[t]]$:*

(1) $R(1 - t - tR) = 1$;
(2) $R - 1 = tR(1 + R)$;
(3) $1 + tR = 2P$;
(4) $P - RP = -tR^2$;
(5) $(1 - tR)(1 + R) = 2$.

*Proof.* Items (1) and (2) are rearrangements of (2.2). By (2), $(1+tR)(1+R) = 1 + R + tR(1+R) = 2R$, which is (3). By (2), $P(1-R) = -R\,tR(1+R)/(1+R) = -tR^2$, which is (4). Finally $(1 - tR)(1 + R) = 1 + R - tR(1 + R) = 2$ by (2). □

2.4. **The recurrences.** We use the following relations; (A), (B) and (C) are due to [3], and (D) is the enumeration of [4].

**Proposition 2.3.** *The numbers $S_{r,k,d}$ satisfy the following relations.*

(A) *For $1 \leq r \leq k$,*

$$S_{r,k,r} = S_{r,k-1,r} + S_{r-1,k-1,r-1} + \sum_{i=2}^{r} \mathrm{Sch}_{i-2}\, S_{r+1-i,k+1-i,r+1-i}.$$

(B) *For $1 \leq k < r$,*

$$S_{r,k,k} = \sum_{j=0}^{k} S_{j,k,j} \binom{r-j-1}{k-j}.$$

(C) *For* $1 \le d < \min(r,k)$,

$$S_{r,k,d} = S_{r,k-1,d} + S_{r-1,k-1,d-1} + \sum_{i=2}^{d} \mathrm{Sch}_{i-2} \left( S_{r+1-i,k+1-i,d+1-i} - \binom{r-i}{d+1-i} \right).$$

(D) *For* $n \ge 1$, $S_{n,n,n} = \mathrm{Sch}_{n-1}$.

*Proof.* Relations (A), (B) and (C) are [3, Proposition 4.2, Proposition 4.3 and Theorem 4.4]. An ASR of size $n \times n$ without empty rows has $n$ rows summing to 1 and $n$ columns summing to at most 1, so it is an ASM. An ASM extendably avoids 312 if and only if it avoids 312: it is an extension of itself, and an occurrence of 312 in it is also one in every extension. Hence (D) is the enumeration of 312-avoiding ASMs in [4]; it is the base case $S_{n,n,n} = \mathrm{Sch}_{n-1}$ used in [3, Section 4]. □

For $r = k$ the term $S_{r,r-1,r}$ in (A) lies outside the range $r, k \ge d \ge 0$ and is zero [3, Section 4]; below, (A) is used only for $r < k$. For (A) with $r < k$, (B) and (C), all arguments of $S$ on the right-hand side lie in the range $r, k \ge d \ge 0$ and have a smaller value of $r + k$ than the left-hand side. The admissible triples $(r, k, d)$ split into the cases $d = 0$, $r = k = d \ge 1$, $1 \le r = d < k$, $1 \le k = d < r$ and $1 \le d < \min(r, k)$, covered by $S_{r,k,0} = 1$, (D), (A), (B) and (C), respectively, so these relations determine $S_{r,k,d}$.

## 3. The diagonal generating functions

Throughout this section $a, b \ge 0$ are integers, $T_{a,b}$ is as in the introduction, and

$$U_a = (1-t)^{-a} = \sum_{h \ge 0} \binom{a+h-1}{h} t^h,$$

by (2.1). Each $T_{a,b}$ has constant term $S_{a,b,0} = 1$.

**Lemma 3.1.** *For* $a \ge 0$ *and* $b \ge 1$,

$$T_{a,b} = R\,T_{a,b-1} - tR^2 U_a.$$

*Proof.* Let $d \ge 1$, $r = d + a$ and $k = d + b$. If $a \ge 1$, then $1 \le d < \min(r, k)$ and (C) applies. If $a = 0$, then $1 \le r < k$ and (A) applies; in this case $\binom{r-i}{d+1-i} = \binom{d-i}{d+1-i} = 0$ for $2 \le i \le d$, so (A) is (C) with the binomial terms omitted. Substituting $j = i - 1$ and $h = d - j$, so that $1 \le j \le d - 1$ and $1 \le h \le d - 1$, we obtain in both cases

$$\begin{aligned} S_{d+a,d+b,d} &= S_{d+a,d+b-1,d} + S_{d-1+a,d-1+b,d-1} \\ &\quad + \sum_{\substack{j+h=d \\ j,h \ge 1}} \mathrm{Sch}_{j-1} \left( S_{h+a,h+b,h} - \binom{a+h-1}{h} \right). \end{aligned} \tag{3.1}$$

The three terms on the right are the coefficients of $t^d$ in $T_{a,b-1}$, in $tT_{a,b}$, and in

$$tR\,(T_{a,b} - 1) - tR\,(U_a - 1),$$

because $tR = \sum_{j \ge 1} \mathrm{Sch}_{j-1} t^j$ and the series $T_{a,b} - 1$ and $U_a - 1$ have zero constant term. For $d = 0$ the coefficient of $t^0$ is 1 on both sides. Hence

$$T_{a,b} = T_{a,b-1} + t(1+R)\,T_{a,b} - tR\,U_a,$$

that is, $(1 - t - tR)\,T_{a,b} = T_{a,b-1} - tRU_a$. Multiplying by $R$ and using Lemma 2.2(1) gives the claim. □

**Lemma 3.2.** *For $a, b \geq 0$,*

$$T_{a,b} = PU_a + R^b\,(T_{a,0} - PU_a).$$

*Proof.* By Lemma 2.2(4), $PU_a = R\,PU_a - tR^2U_a$. Subtracting this from Lemma 3.1 gives $T_{a,b} - PU_a = R\,(T_{a,b-1} - PU_a)$ for $b \geq 1$, and the claim follows by induction on $b$. □

**Lemma 3.3.** *For $b \geq 0$, $T_{0,b} = P\,(1 + R^b)$.*

*Proof.* By Proposition 2.3(D), $T_{0,0} = 1 + \sum_{d\geq 1} \mathrm{Sch}_{d-1} t^d = 1 + tR = 2P$, using Lemma 2.2(3). Since $U_0 = 1$, Lemma 3.2 gives $T_{0,b} = P + R^b(2P - P)$. □

**Lemma 3.4.** *For $a \geq 0$,*

$$T_{a,0} = P\left((1-t)^{-a} + (1-tR)^{-a}\right).$$

*Proof.* For $a = 0$ this is $T_{0,0} = 2P$. Let $a \geq 1$. We claim that for every $d \geq 0$

$$S_{d+a,d,d} = \sum_{h=0}^{d} \binom{a+h-1}{h} S_{d-h,d,d-h}. \tag{3.2}$$

For $d = 0$ both sides equal 1. For $d \geq 1$ this is (B) with $r = d + a$, $k = d$ and $j = d - h$, since $\binom{r-j-1}{k-j} = \binom{a+h-1}{h}$. Now $S_{d-h,d,d-h}$ is the coefficient of $t^{d-h}$ in $T_{0,h}$, because $(d-h, d, d-h) = (e, e+h, e)$ with $e = d - h$. Therefore (3.2) says that

$$T_{a,0} = \sum_{h\geq 0} \binom{a+h-1}{h} t^h\, T_{0,h},$$

where the sum is defined because $t^h T_{0,h} \in t^h\mathbb{Q}[[t]]$. By Lemma 3.3 and (2.1), applied with $x = t$ and with $x = tR$,

$$T_{a,0} = P \sum_{h\geq 0} \binom{a+h-1}{h} \left(t^h + (tR)^h\right) = P\left((1-t)^{-a} + (1-tR)^{-a}\right). \qquad \square$$

*Proof of Theorem 1.2.* By Lemma 3.4, $T_{a,0} - PU_a = P(1-tR)^{-a}$. Inserting this into Lemma 3.2 gives $T_{a,b} = P(1-t)^{-a} + PR^b(1-tR)^{-a}$. □

*Proof of Corollary 1.3.* For fixed $a$ and $b$ the coefficient of $x^a y^b$ on the left is $T_{a,b}$. On the right, expanding the geometric series in $x$ and $y$, the coefficient of $x^a y^b$ is $P(1-t)^{-a} + PR^b(1-tR)^{-a}$. The two sides agree by Theorem 1.2. By (2.3), $R$ and $P$ lie in $\mathbb{Q}(t)\big(\sqrt{1-6t+t^2}\big)$, so the right-hand side lies in the field $\mathbb{Q}(x,y,t)\big(\sqrt{1-6t+t^2}\big)$, which has degree at most two over $\mathbb{Q}(x,y,t)$. □

## 4. The generating function of the conjectured formula

For integers $r, k \geq 0$ we regard (1.1) as defining $F_{r,k,d}$ for every $d \geq 0$, and put

$$G_{r,k}(z) = \sum_{d\geq 0} F_{r,k,d}\, z^d.$$

The values with $d > \min(r,k)$ are used only as coefficients of this auxiliary series; they carry no combinatorial meaning. Let

$$C(u) = \sum_{j\geq 0} C_j u^j, \qquad E(u) = \frac{C(u) + C(-u)}{2} = \sum_{i\geq 0} C_{2i} u^{2i}.$$

**Lemma 4.1.** *For integers $r, k \geq 0$,*

$$G_{r,k}(z) = \tfrac{1}{2}(1+z)^{r-2}\left((1-2z^2)(1+2z)^k + 1\right) + z(1+z)^{r-1}E\big(z(1+z)\big).$$

*Proof.* We compute the generating functions of the four summands in (1.1) separately. In the first summand put $m = d - i \geq 0$. Since $\binom{r-2}{m-2} = 0$ for $m < 2$,

$$\sum_{d\geq 0} z^d \sum_{i=1}^{d} 2^{i-1}\binom{k}{i}\left(\binom{r-2}{d-i} - 2\binom{r-2}{d-i-2}\right) = \frac{(1+2z)^k - 1}{2}\,(1-2z^2)(1+z)^{r-2}.$$

The second summand gives $(1+z)^r$. In the third summand put $m = d - 1 - 2i$; then

$$\sum_{d\geq 0} z^d \sum_{i\geq 0} C_{2i}\binom{r-1+2i}{d-1-2i} = \sum_{i\geq 0} C_{2i}\, z^{1+2i}(1+z)^{r-1+2i} = z(1+z)^{r-1}E\big(z(1+z)\big),$$

where the sum over $i$ is defined because of the factor $z^{1+2i}$. In the last summand put $m = d - i$; then

$$2\sum_{d\geq 0} z^d \sum_{i=1}^{d}(-1)^i\binom{r}{d-i} = 2(1+z)^r \sum_{i\geq 1}(-z)^i = -2z(1+z)^{r-1}.$$

Finally $(1+z)^r - 2z(1+z)^{r-1} = (1+z)^{r-2}(1-z^2)$, and $-\frac{1}{2}(1-2z^2) + (1-z^2) = \frac{1}{2}$. Adding the four contributions gives the stated formula. □

We use the following standard diagonal form of Lagrange inversion. For a formal Laurent series $L(z) = \sum_{n\geq n_0} \ell_n z^n$ we write $\operatorname{Res}_z L = \ell_{-1}$.

**Lemma 4.2.** *Let $\varphi, H \in \mathbb{Q}[[z]]$ with $\varphi(0) = 1$, and let $w \in t\mathbb{Q}[[t]]$ be the unique series with $w = t\,\varphi(w)$. Then $1 - t\varphi'(w)$ is invertible and*

$$\sum_{n\geq 0}\left([z^n]\,H(z)\varphi(z)^n\right)t^n = \frac{H(w)}{1 - t\varphi'(w)}.$$

*Proof.* The coefficients of $w$ are determined recursively by $w = t\varphi(w)$, and $1 - t\varphi'(w)$ has constant term 1. Let $g(z) = z/\varphi(z)$, so that $g(0) = 0$ and $g'(0) = 1$. The relation $w = t\varphi(w)$ says that $g(w(t)) = t$, so $w$ is the compositional inverse of $g$ and $w(g(z)) = z$.

We use the change of variables formula $\operatorname{Res}_t L(t) = \operatorname{Res}_z L(g(z))\,g'(z)$, valid for every formal Laurent series $L$. Terms of $L$ with nonnegative exponent contribute power series on both sides, so by linearity it suffices to check the formula for $L(t) = t^m$ with $m < 0$. For $m \leq -2$ the series $g^m g' = (g^{m+1})'/(m+1)$ is a derivative and has residue 0. For $m = -1$, writing $g = zu$ with $u(0) = 1$, we have $g'/g = 1/z + u'/u$, whose residue is 1.

Let $A(t) = H(w)/(1 - t\varphi'(w))$ and $n \geq 0$. Then $A(g(z)) = H(z)/\big(1 - g(z)\varphi'(z)\big)$ and

$$[t^n]\,A = \operatorname{Res}_t A(t)\,t^{-n-1} = \operatorname{Res}_z \frac{H(z)\,g'(z)}{1 - g(z)\varphi'(z)}\,g(z)^{-n-1}.$$

Since $g' = (\varphi - z\varphi')/\varphi^2$ and $1 - g\varphi' = (\varphi - z\varphi')/\varphi$, where $\varphi - z\varphi'$ has constant term 1, we get $g'/(1 - g\varphi') = 1/\varphi$. As $g^{-n-1} = \varphi^{n+1}z^{-n-1}$, the residue equals $\operatorname{Res}_z H(z)\varphi(z)^n z^{-n-1} = [z^n]\,H(z)\varphi(z)^n$. □

We also need the even part of the Catalan series at a particular argument.

**Lemma 4.3.** *With $u = t/(1-t)^2$ we have $E(u) = \frac{1}{2}(1-t)(1+R)$.*

*Proof.* The Catalan series satisfies $1-2uC(u) = \sqrt{1-4u}$, hence also $1+2uC(-u) = \sqrt{1+4u}$. Therefore

$$4u\,E(u) = 2uC(u) + 2uC(-u) = \sqrt{1+4u} - \sqrt{1-4u}.$$

Substitute $u = t/(1-t)^2$, which has zero constant term. Then $1+4u = (1+t)^2/(1-t)^2$ and $1-4u = (1-6t+t^2)/(1-t)^2$. By uniqueness of square roots with constant term 1, the substituted square roots are $(1-t)^{-1}(1+t)$ and $(1-t)^{-1}\sqrt{1-6t+t^2}$. Using (2.3),

$$\frac{4t}{(1-t)^2}\,E(u) = \frac{1+t-\sqrt{1-6t+t^2}}{1-t} = \frac{2t(1+R)}{1-t},$$

and cancelling $2t$, which is possible since $\mathbb{Q}[[t]]$ is an integral domain, gives the claim. □

**Proposition 4.4.** *For all integers $a, b \geq 0$,*

$$\sum_{d\geq 0} F_{d+a,d+b,d}\,t^d = P\left(\frac{1}{(1-t)^a} + \frac{R^b}{(1-tR)^a}\right).$$

*Proof.* By definition $F_{d+a,d+b,d} = [z^d]\,G_{d+a,d+b}(z)$. By Lemma 4.1, with $r = d+a$ and $k = d+b$,

$$G_{d+a,d+b}(z) = H_1(z)\varphi_1(z)^d + H_2(z)\varphi_2(z)^d,$$

where

$$\varphi_1 = (1+z)(1+2z), \qquad H_1 = \tfrac{1}{2}(1+z)^{a-2}(1-2z^2)(1+2z)^b,$$
$$\varphi_2 = 1+z, \qquad H_2 = \tfrac{1}{2}(1+z)^{a-2} + z(1+z)^{a-1}E\big(z(1+z)\big).$$

These are power series in $z$, and $\varphi_1(0) = \varphi_2(0) = 1$. We apply Lemma 4.2 to both terms.

For $\varphi_1$, put $w_1 = (R-1)/2 \in t\mathbb{Q}[[t]]$. By Lemma 2.2(2),

$$t\varphi_1(w_1) = t\cdot\frac{1+R}{2}\cdot R = \frac{R-1}{2} = w_1,$$

so $w_1$ is the series of Lemma 4.2. Moreover $1+2w_1 = R$ and, by Lemma 2.2(5), $1+w_1 = (1+R)/2 = (1-tR)^{-1}$. Since $\varphi_1'(z) = 3+4z$ and $t = w_1/\big((1+w_1)(1+2w_1)\big)$,

$$1 - t\varphi_1'(w_1) = \frac{(1+w_1)(1+2w_1) - w_1(3+4w_1)}{(1+w_1)(1+2w_1)} = \frac{1-2w_1^2}{(1+w_1)(1+2w_1)}.$$

The factor $1-2w_1^2$ has constant term 1 and cancels against the factor $1-2z^2$ of $H_1$ evaluated at $w_1$. Hence

$$\frac{H_1(w_1)}{1-t\varphi_1'(w_1)} = \tfrac{1}{2}(1+w_1)^{a-1}(1+2w_1)^{b+1} = \tfrac{1}{2}(1-tR)\,R\cdot\frac{R^b}{(1-tR)^a} = \frac{PR^b}{(1-tR)^a},$$

where the last step uses $\frac{1}{2}(1-tR)R = R/(1+R) = P$, by Lemma 2.2(5).

For $\varphi_2$, the series $w_2 = t/(1-t)$ satisfies $w_2 = t(1+w_2)$, and $1+w_2 = (1-t)^{-1}$, $w_2(1+w_2) = t/(1-t)^2$ and $1 - t\varphi_2'(w_2) = 1-t$. Hence, by Lemma 4.3,

$$\frac{H_2(w_2)}{1-t} = \tfrac{1}{2}(1-t)^{1-a} + \frac{t}{(1-t)^{a+1}}\,E\!\left(\frac{t}{(1-t)^2}\right) = \tfrac{1}{2}(1-t)^{1-a} + \tfrac{1}{2}\,t\,(1-t)^{-a}(1+R).$$

This equals $\frac{1}{2}(1-t)^{-a}(1+tR) = P(1-t)^{-a}$ by Lemma 2.2(3).

By Lemma 4.2 and linearity of coefficient extraction, $\sum_d F_{d+a,d+b,d}t^d$ is the sum of the two contributions, which is the asserted series. □

## 5. Proof of the main theorem

*Proof of Theorem 1.1.* Let $r, k \geq d \geq 0$, and put $a = r - d$ and $b = k - d$, which are nonnegative integers. By Theorem 1.2 and Proposition 4.4, the series $\sum_{e\geq 0} S_{e+a,e+b,e}t^e$ and $\sum_{e\geq 0} F_{e+a,e+b,e}t^e$ coincide. Comparing the coefficients of $t^d$ gives $S_{r,k,d} = F_{r,k,d}$. □

*Remark* 5.1. Theorem 1.2 makes some special cases explicit. For $a = b = 0$ it gives $\sum_{d\geq 0} S_{d,d,d}t^d = 2P = 1+tR$, so Proposition 4.4 yields a second proof of the identity $F_{n,n,n} = \mathrm{Sch}_{n-1}$ of [3, p. 9]. For $a = 0$ and $b = 1$ it gives $T_{0,1} = P(1+R) = R$, that is, $S_{d,d+1,d} = \mathrm{Sch}_d$ for all $d \geq 0$. For small $d$, Theorem 1.1 and a direct evaluation of (1.1) give

$$S_{r,k,1} = r + k - 1 \quad (r, k \geq 1), \qquad S_{r,k,2} = k(k + r - 3) + \binom{r-1}{2} \quad (r, k \geq 2).$$

Table 1 lists the values for $d = 3$.

Table 1. The numbers $S_{r,k,3}$ for $3 \leq r, k \leq 8$.

| $r\backslash k$ | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|
| 3 | 6 | 22 | 52 | 100 | 170 | 266 |
| 4 | 16 | 39 | 78 | 137 | 220 | 331 |
| 5 | 31 | 62 | 111 | 182 | 279 | 406 |
| 6 | 52 | 92 | 152 | 236 | 348 | 492 |
| 7 | 80 | 130 | 202 | 300 | 428 | 590 |
| 8 | 116 | 177 | 262 | 375 | 520 | 701 |

*Remark* 5.2. We have also checked the ingredients of the proof numerically, using exact integer and rational arithmetic. The values of $F_{r,k,d}$ agree with those computed from Proposition 2.3 for all 10,416 triples with $0 \leq r, k \leq 30$ and $0 \leq d \leq \min(r, k)$, and the coefficients of $t^d$ in Theorem 1.2 agree with $F_{d+a,d+b,d}$ for $0 \leq a, b \leq 8$ and $0 \leq d \leq 20$. Without using the recurrences, we generated all ASMs of orders 1 to 6 directly from monotone triangles; there are $1, 2, 7, 42, 429, 7436$ of them, of which $1, 2, 6, 22, 90, 394$ avoid 312, in accordance with Proposition 2.3(D). For each of the 30 admissible triples $(r, k, d)$ with $r, k \leq 3$, the number of distinct upper left $r \times k$ corners of these 312-avoiding ASMs that are ASRs with exactly $d$ nonempty rows equals $F_{r,k,d}$; the restriction to ASRs is essential, since a row of such a corner may end with $-1$. This last check exhibits extensions of order at most six for the counted rectangles; it is not an exhaustive search over extensions of arbitrary order and is not used in the proof.

## Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

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